

Dimitri Callegari Jadovski

Backend & Fullstack Developer | Software Engineering Intern @ Banrisul | BS Computer Science, PUCRS
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SUMMARY

Computer Science student and backend-focused developer. Writes C# in production on an internal AI assistant at one of Brazil's largest regional banks, and TypeScript on personal projects — a 14,500-line investment analytics platform with 273 tests, PostgreSQL and custom session auth, and a 30,500-line server-authoritative multiplayer game. Builds systems end to end: designs the data model, writes the domain logic, tests it against a real database in CI, and ships it.

EXPERIENCE

Banrisul — Fullstack Developer Intern

2025 – Present | Porto Alegre, RS | C#, JavaScript, HTML5, CSS3, Microsoft Copilot Studio

- Work across the stack on **BanriChat**, an internal AI assistant used by bank employees, from agent configuration through the C# backend to the browser client.
- Curate the knowledge sources behind the assistant: extract documentation from the bank's internal wiki and load it into **Microsoft Copilot Studio** agents scoped by subject domain, so each agent answers from a deliberately bounded corpus.
- Consume those agents from the C# backend, routing each incoming request to the agent that matches its scope and handling latency, failures, and unexpected responses so the assistant behaves predictably for end users.
- Build the client interface in vanilla JavaScript, HTML5 and CSS3, wiring it to backend endpoints over HTTP and applying RESTful API principles.
- Collaborate within a corporate engineering team focused on operational efficiency.

PROJECTS

PortfolioOS — Investment Portfolio Analytics Platform

TypeScript, Next.js, PostgreSQL, Prisma, Vitest, GitHub Actions | github.com/Dimigol/PortfolioOS

- Architected a **~14,500-line** modular monolith with a strict service layer (analytics, auth, imports, benchmarks, snapshots), backed by a 13-table PostgreSQL schema managed through versioned Prisma migrations.
- Implemented **XIRR** (internal rate of return over irregular cash flows) via Newton–Raphson iteration, plus Brazilian regressive income-tax estimation per Lei 11.033/2004 — deliberately biased toward over-estimating tax so gross-yield comparisons between taxed and tax-exempt instruments stay honest.
- Built **session authentication from scratch**: bcrypt password hashing ($N=2^{16}$, tuned to cap memory per login attempt), constant-time comparison, and a two-layer sliding-window rate limiter keyed on account+IP and IP separately to close the credential-stuffing gap that per-account limiting alone leaves open.
- Wrote a **coordinate-based PDF table extractor** that reconstructs broker statements by clustering text fragments on the Y axis and sorting on X — deterministic, where naive text extraction interleaves columns and corrupts values.
- Integrated external market data from the Banco Central **SGS API** (CDI, SELIC, IPCA, USD/BRL) and Yahoo Finance, with a scheduled daily job that revalues positions and writes historical snapshots.
- Maintained **273 automated tests** run against a real PostgreSQL 18 instance in CI — chosen over mocks so constraints, cascades, and transaction behavior are actually exercised — gated by typecheck, migration, and build stages in GitHub Actions.

Steal a Startup — Server-Authoritative Multiplayer Game

Luau, Roblox, Rojo, Wally, Selene | github.com/Dimigol/Steal-a-Startup

- Designed and built a **~30,500-line** multiplayer economy game across 20+ decoupled server services (economy, employees, theft, security, rebirth, monetization, analytics) communicating through an event bus rather than direct calls.
- Enforced a **server-authoritative** model in which the client only requests and renders: every currency, purchase, and theft resolution is computed server-side, with typed remote contracts, validation middleware, and per-endpoint rate limiting on all client-to-server traffic.
- Isolated persistence behind a **provider contract validated at boot**, allowing in-memory and DataStore backends to be swapped by configuration without touching any gameplay service.

- Extracted all balancing constants into **23 dedicated config modules**, so economy tuning never requires editing logic, and documented the architecture, persistence strategy, and balancing model in maintained design docs.
- Covered core systems (theft probability math, income caps, employee progression, rebirth reset, wallet) with a unit test suite, enforced by Selene linting and StyLua formatting.

Coinverter — Currency & Cryptocurrency Converter

JavaScript, HTML5, CSS3, Chart.js | github.com/Dimigol/Coinverter

- Built a zero-dependency, no-build-step web app consuming two public REST APIs: **AwesomeAPI** for fiat conversion and **CoinGecko** for the top 20 cryptocurrencies.
- Rendered interactive price-history charts (24h to 1y) with Chart.js, handling asynchronous fetches, rate-limit responses, and error states without a framework.

Tic-Tac-Toe with AI

JavaScript

- Implemented the **Minimax** algorithm with depth-limited search to produce an unbeatable opponent, with difficulty levels mapped to search depth.

SKILLS

Languages: C#, TypeScript, JavaScript, Java, Python, Luau

Backend: Node.js, Next.js, FastAPI, HTTP, RESTful APIs, SQL, PostgreSQL, Prisma, Docker, Zod

AI Integration: Microsoft Copilot Studio, OpenAI API, knowledge-base agent design

Testing & CI: Vitest, GitHub Actions, integration testing against live databases

Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Relational Data Modeling, Service Architecture, Authentication & Rate Limiting

Frontend: React, HTML5, CSS3, Tailwind CSS, Chart.js

Tools: Git, GitHub, VS Code, IntelliJ IDEA

EDUCATION

PUCRS — Pontifical Catholic University of Rio Grande do Sul

Bachelor of Science in Computer Science | 2025 – 2028 (expected)

CERTIFICATIONS

- OpenAI and Python: Building Chatbots with the OpenAI API — Alura (48h)
- Object-Oriented Python — Alura (25h)
- Entrepreneurship for Developers — Alura (76h)
- Programming Fundamentals — Alura (31h)

ACHIEVEMENTS

- **1st Place** — Marista Robotics Festival, Drone Challenge (Advanced Category)
- Young Scientist Space, PUCRS (2018) — Project: *Logic Codes*
- UFRGS Young Science Fair, UFRGS (2018) — Project: *Logic Codes*
- Robotics and Programming Club, CMR (2022–2023)

LANGUAGES

Portuguese — Native | **English** — Advanced | **Spanish** — Intermediate